

SLIP – 8

Section I: Data Structures [15 Marks]

Q1) Sort a random array of n integers (accept the value of n from user) in ascending order by using bubble sort algorithm.

Answer:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int a[100], n, i, j, temp;

    printf("Enter n: ");
    scanf("%d", &n);

    if(n <= 0 || n > 100)
    {
        printf("n must be between 1 and 100");
        return 0;
    }

    for(i = 0; i < n; i++)
        a[i] = rand() % 100;

    printf("Original array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp
= a[j];
                a[j] = a[j
+ 1];
                a[j + 1] =
temp;
            }
        }
    }
}
```

```

        printf("\nSorted array: ");
for(i = 0; i < n; i++)
printf("%d ", a[i]);

        return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following Student-Teacher database(primary keys are underlined).

Student (sno , s_name, s_class)

Teacher (tno , t_name, yrs_experience)

The relationship is as follows:

Student-Teacher: Many to Many with descriptive attribute Subject.

Constraint:s_class should be either "FY", "SY" or "TY"

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Student(
sno INT PRIMARY KEY,
s_name VARCHAR(50),
s_class VARCHAR(2)
        CHECK(s_class IN ('FY','SY','TY'))
);

CREATE TABLE Teacher(
tno INT PRIMARY KEY,
t_name VARCHAR(50),
yrs_experience INT
);

CREATE TABLE Student_Teacher(
        sno INT REFERENCES Student(sno) ON DELETE CASCADE,
tno INT REFERENCES Teacher(tno) ON DELETE CASCADE,
subject VARCHAR(50),
        PRIMARY KEY(sno, tno, subject)
);

```

```
INSERT INTO Student VALUES
(101, 'Amit', 'FY'),
(102, 'Neha', 'SY'), (103, 'Raj', 'TY'),
(105, 'Sita', 'TY');
```

```
INSERT INTO Teacher VALUES
(1, 'Pawar', 8),
(2, 'Joshi', 3),
(3, 'Kale', 12);
```

```
INSERT INTO Student_Teacher VALUES
(101, 1, 'Maths'),
(102, 2, 'Data Structure'),
(103, 1, 'Data Structure'),
(105, 3, 'Maths'),
(103, 3, 'DBMS');
```

Q 2(B)

B. Execute the following queries in PostgreSQL (any 3) [6 marks]

i. Give class-wise number of students.

Answer:

```
SELECT s_class, COUNT(*)
FROM Student
GROUP BY s_class; ii. List all students
studying in class "TY".
```

Answer:

```
SELECT * FROM Student
WHERE s_class = 'TY';
```

iii. Count the number of students who have taken subject "Maths"

Answer:

```
SELECT COUNT(DISTINCT sno)
FROM Student_Teacher
WHERE subject = 'Maths';
```

iv. Delete record of student whose sno "105"

Q 2(C)

C. Write a query to list the names of all teachers with their subjects along with the total number of students they are teaching. [4 marks]

Answer:

```
SELECT T.t_name, ST.subject,  
       COUNT(DISTINCT ST.sno) AS total_students  
FROM Teacher T  
JOIN Student_Teacher ST ON T.tno = ST.tno  
GROUP BY T.t_name, ST.subject;
```

Q 3) Viva [5 marks]